

SURENDRA KUMAR REDDY RUKA

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🐙 GitHub

in LinkedIn

🌐 Portfolio

Objective

Enthusiastic Embedded Systems Engineer with hands-on experience in STM32-based firmware, real-time communication systems, and hardware design. Skilled in C, C++, and embedded driver development for industrial and IoT applications. Adept at creating robust, efficient, and scalable embedded solutions while ensuring performance optimization and system reliability.

Skills

Programming :	C, C++, Embedded C, Python
Microcontrollers :	STM32 (F1/F3/F4/F7 series), Arduino, ESP32
Protocols :	UART, SPI, I2C, RS485, CAN, Modbus RTU/TCP, MQTT, Bluetooth, LoRa
Embedded Systems :	FreeRTOS, Device Drivers, Interrupts, Timers, ADC/DAC, Sensor Integration
PCB Tools :	KiCad, Altium Designer, DipTrace, Eagle
Dev Tools :	STM32CubeIDE, STM32CubeProgrammer, Arduino IDE, Git, GitHub
Other :	SCADA, DWIN GUI

Experience

Embedded System Engineer

Radiogeect Digital Pvt Ltd – Jamshedpur, India

April 2024 – Present

- Developed **Radiowave Remote IO**, a wireless P2P transceiver system using LoRa with a range of 10 km.
- Engineered bidirectional communication for analog and digital data transmission using **STM32F407**.
- Designed a 4-channel process display system utilizing **STM32F373** with real-time visualization.
- Integrated **ADS1115** for analog input acquisition and **MCP4725** for 4–20 mA output transmission.
- Built an energy-saving system using **LD1115H radar** for human presence monitoring.
- Designed a **UWB-based position tracking system** for EOT cranes using STM32 + FreeRTOS.
- Developed a **radar-based vehicle speed monitoring system** using LD2415H + ESP32 with P4 LED matrix.

Embedded System Developer Intern

Radiogeect Digital Pvt Ltd – Jamshedpur, India

July 2023 – March 2024

- Developed firmware for a process controller using **STM32F373C8T6**, integrating **ADS1115** and **MAX485**.
 - Designed PCBs for process controllers with 4–20 mA / 1–5 V inputs and 4-inch display.
 - Built custom libraries for **LED matrix displays** (P10, P4, P2.5) on STM32F407VGT6.
 - Designed RGB and single-color LED matrix driver PCBs using **STM32F103C8T6**.
 - Developed RS485-based temperature/humidity display using **SHT20 sensors**.
 - Generated 4–20 mA analog outputs from sensor data.
 - Built anemometer display by interfacing wind speed sensors with STM32F373C8T6.
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Education

B.Tech – Electronics & Communication Engineering

Rajiv Gandhi University of Knowledge Technologies, Srikakulam
GPA: 8.46

April 2024

Pre-University Course – MPC

Rajiv Gandhi University of Knowledge Technologies, Srikakulam
GPA: 8.0

May 2020

Class X – SSC Board

Zilla Parishad High School, R.V. Palli
GPA: 9.8

April 2018

Academic Projects

Amplifying Education System with Real-Time Updates – STM32, LoRa, GSM

Wireless classroom communication system enabling real-time announcements and automated timetables.

Accident Prevention System – Arduino, Eye-Blink Sensor

Driver drowsiness detection system based on eye-blink monitoring.

Home Automation System – ESP32, Blynk

IoT-based appliance control via ESP32 using Blynk mobile application.

STM32F405 Eval Board Design – KiCad, 4-layer PCB

Designed a custom STM32F405 development board supporting UART, SPI, I2C, USB, and GPIO.

Accomplishments

- **State-Level Chess Player** and 3-time **District First Place** winner.
 - **Udemy Certifications:**
 - Mastering Microcontroller and Embedded Driver Development (MCU1)
 - Mastering Microcontroller: Timers, PWM, CAN, Low Power (MCU2)
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Languages

Telugu : Native
English : Intermediate
Hindi : Intermediate